



Allogeneic hematopoietic stem cell transplantation (allo-HCT) remains a key curative treatment for acute myeloid leukemia (AML)

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Summary

Introduction This review summarizes the current state of evidence from peer-reviewed publications and relevant conference presentations from 2023–2025 and highlights the impact of patient selection, donor selection, conditioning, measures to prevent graft-versus-host disease (GvHD), maintenance therapy, monitoring of measurable residual disease (MRD), and supportive measures on the transplant outcome.

Methods Narrative review with current randomized, large retrospective studies and important abstracts on adult acute myeloid leukemia (AML) patients who have undergone stem cell transplantation (allo-HCT). Conference contributions are included in the evaluation if mature data are still lacking evaluation.

Results We see the greatest benefit of allo-HCT in patients with unfavorable genetics or MRD persistence after initial complete remission. MRD-negative patients with favorable genetics achieve long-term cure, in some cases even without transplantation [1, 2]. In several studies, haploidentical transplants with post-transplant cyclophosphamide (PTCy) have achieved results similar to those of HLA-identical siblings or unrelated donors; however, registry data report higher nonrelapse mortality (NRM) or poorer overall survival (OS) in individual settings. In mismatched unrelated donor transplantation, PTCy was superior to rabbit antithymocyte globulin (rATG) for NRM and OS. With iomab-B (radioimmunological approach), older pa-

tients with relapsed/refractory AML can receive targeted curative treatment. Sorafenib as maintenance therapy after allo-HCT improves relapse-free survival and OS in FLT3-ITD-mutated AML, while a benefit from gilteritinib after allo-HCT was particularly shown for MRD-positive patients. APR-246 (eprenetapopt): Phase II data support its use as maintenance therapy after allo-HCT and early pilot studies with enasidenib or ivosidenib report good tolerability and efficacy in small cohorts. MRD monitoring after HCT identifies molecular relapse and patients at high risk for overt relapse who may benefit from pre-emptive interventions; letermovir reduces CMV disease (not reactivation) to <4%, higher gut microbiome diversity correlates with lower GvHD risk. Whether prophylactic fecal microbiota transfer (FMT) may reduce GvHD remains unproven in randomized trials.

Conclusion In addition to ELN risk criteria and MRD status for accurate risk stratification, the wider availability of alternative donors thanks to PTCy-based GvHD prophylaxis, targeted maintenance therapy, and improved supportive care increase the curative potential of allo-HCT in AML.

Keywords Measurable residual disease · Graft-versus-host disease · Overall survival · Maintenance therapy · Risk stratification

Abbreviations

aGVHD	Acute graft-versus-host disease
Allo-HCT	Allogeneic hematopoietic stem cell transplantation
Allo-HSCT	Allogeneic hematopoietic stem cell transplantation
AML	Acute myeloid leukemia
AML/MDS	Acute myeloid leukemia/myelodysplastic syndromes
APR-246	Eprenetapopt (TP53-reactivating agent)

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ATG	Antithymocyte globulin
BMT CTN	Blood and Marrow Transplant Clinical Trials Network
cGVHD	Chronic graft-versus-host disease
CI	Confidence interval
CNI	Calcineurin inhibitor
CR1	First complete remission
DLI	Donor lymphocyte infusion
CNI/MTX	Calcineurin inhibitor/methotrexate
CTLA-4-Ig	Cytotoxic T-lymphocyte-associated protein 4-immunoglobulin fusion protein
ELN	European LeukemiaNet
GI	Gastrointestinal
GI-aGVHD	Gastrointestinal acute graft-versus-host disease
GRFS	GvHD-free, relapse-free survival
GvHD	Graft-versus-host disease
GvL	Graft-versus-leukemia
HCT	Hematopoietic cell transplantation
HLA	Human leukocyte antigen
HMA	Hypomethylating agent(s)
HR	Hazard ratio
HSCT	Hematopoietic stem cell transplantation
IAT	Investigator-assigned therapy
IDH1	Isocitrate dehydrogenase 1
IDH2	Isocitrate dehydrogenase 2
Iomab-B	131I-apamistamab (anti-CD45 radioimmunoconjugate)
MAC	Myeloablative conditioning
MMUD	Mismatched unrelated donor
MRD	Measurable residual disease
MSD	Matched sibling donor
MTX	Methotrexate
MUD	Matched unrelated donor
NRM	Nonrelapse mortality
OS	Overall survival
PTCy	Posttransplant cyclophosphamide
rATG	Rabbit antithymocyte globulin
RCT	Randomized controlled trial
RFS	Relapse-free survival
RIC	Reduced-intensity conditioning
RIC/MAC	Reduced-intensity/myeloablative conditioning
Tac	Tacrolimus
Tac/MMF	Tacrolimus/mycophenolate mofetil
TKI	Tyrosine kinase inhibitor
Tregs	Regulatory T cells
Ven/Aza	Venetoclax plus azacitidine

Take home message

- Allo-HCT offers the best curative chance in the majority of nonfavorable risk AML patients
- MRD- and genetics-guided transplant indication
- Expanded donor availability due to PTCy, including haploidentical and mismatched unrelated donors
- MRD-guided post-HCT pre-emptive strategies to prevent relapse

Introduction

By introducing a potent graft-versus-leukemia mechanism, allo-HCT offers a chance of cure in AML. The adverse effect of this immunotherapy approach, the frequent occurrence of graft-versus-host disease, is the main cause for a significant rate of nonrelapse mortality. Even more frequently, recurrence of leukemia may hamper the success of this treatment approach. Nevertheless, allo-HCT provides the highest chance for long-term disease control for the majority of AML patients who are eligible for this approach.

Moreover, the above limitations are being overcome step by step over the years through better risk classification, more open transplant options, reduced GvHD rates, and relapse prevention and treatment. Today, allo-HCT is an option even for many elderly patients. The decision should be made on a case-by-case basis, taking into account genetic risks, MRD status, comorbidities, new conditioning regimens, GvHD prophylaxis, and follow-up therapies in order to maximize benefit and minimize toxicity.

Indication: ELN risk and MRD

In the ELN risk classifier 2022, AML is divided into favorable, intermediate, and unfavorable risk groups based on cytogenetic and molecular findings [1]. Patients with unfavorable ELN risk or persistent MRD after induction should generally receive allo-HSCT. Favorable-risk patients (e.g., RUNX1-RUNX1T1 or NPM1 without unfavorable co-mutations or FLT3-ITD, which is always classified as intermediate in ELN-2022) can often be cured with conventional consolidation if MRD-negative and do not require a transplant in first complete remission. MRD status differentiates the recommendations for intermediate risk AML: In a British evaluation (AML 17/19, reported as a conference abstract), MRD-positive *NPM1*-mutated patients had a significant survival advantage with allo-HSCT (3-year OS 61% with vs. 24% without allo-HCT), whereas MRD-negative patients showed no benefit (3-year OS 75%/83% in AML17/AML19; no additional benefit from allo-HCT, regardless of *FLT3-ITD* co-mutations) [2].

Complementing these data, a randomized phase III trial in patients ≤60 years with intermediate-risk AML in first remission found that primary allo-HCT vs standard consolidation chemotherapy improved disease-free survival and reduced relapse, but did not improve overall survival; nearly all patients in the chemotherapy arm underwent salvage transplantation at relapse [3]. However, it remains questionable whether this favorable outcome of relapsed AML is reproducible in the real-world setting, particularly when the ELN instead of the MRC risk stratification is applied in which a proportion of MRC intermediate-risk patients are assigned to the favorable risk group. MRD may there-

Table 1 Proposed indications for allo-HCT in CR1 according to ELN risk and MRD

ELN-risk group (examples)	MRD status	Recommendation in CR1	Evidence (examples)
Favorable (e.g., NPM1 without FLT3-ITD, RUNX1-RUNX1T1, CBFB::MYH11, bZIP in-frame mutated CEBPA)	MRD negative	No HCT in CR1	MRD-negative patients do not gain a survival benefit from HCT [2]
Favorable/intermediate	MRD-positive	HCT in CR1	MRD-positive cases show markedly improved survival with HCT [2]
Intermediate (e.g., FLT3-ITD with low allelic burden)	MRD-negative	Depends on age, comorbidity, donor	Data are heterogeneous; some MRD-negative patients may forgo HCT [1]
Intermediate	MRD-positive	HCT	MRD persistence is associated with high relapse risk [2]
Adverse (e.g., TP53 mutation, complex karyotype)	Any	HCT in CR1	High relapse rate makes HCT necessary [1]

fore serve as a gatekeeper for transplant indication, at least in MRC favorable or intermediate-risk patients [4]. AML with unfavorable genetics, especially *TP53* mutation or complex karyotype, remains a clear argument for transplantation in CR1 ([1]; Table 1).

Donor selection and transplantation procedure

Haploidentical donors and PTCy The administration of high-dose cyclophosphamide on days +3 and +4 after transplantation has fundamentally changed transplantation from single-haplotype-matched family donors. PTCy induces functional dysfunction/preservation of Tregs and prevents severe GvHD without compromising engraftment velocity [5]. Registry data show that overall survival, relapse risk, and non-relapse mortality with haploidentical use of PTCy are roughly equivalent to those of fully HLA-matched sibling or unrelated donor transplants [6]. In mismatched unrelated donor transplants, PTCy reduces nonrelapse mortality (HR 0.74) and improves overall survival (HR 0.77) compared to ATG treatment [7].

Fully HLA-identical sibling and unrelated donors

A fully HLA-identical sibling donor remains the first choice, if available; HLA-identical unrelated donors achieve comparable results but require a longer [8] search. In younger patients, a young MUD may be preferable to an older haploidentical donor [9]. Criteria such as donor age, sex, CMV serostatus, and blood type influence the decision. The addition of abatacept (CTLA-4-Ig) to tacrolimus/methotrexate reduces the risk of severe acute GvHD in 7/8 mismatch transplants [10].

Umbilical cord blood and mismatched donors

Umbilical cord blood transplants offer immediate availability and a lower risk of relapse, but are associated with delayed engraftment and higher NRM. For 7/8 MMUD with abatacept + CNI/MTX, survival/GRFS was shown to be comparable to 8/8 MUD; for PTCy vs. rATG, there are no consistent advantages in NRM/OS. Overall, an acceptable donor is now available for almost all patients.

Conditioning and GvHD prophylaxis

Radioimmunotherapy with Iomab-B Radioimmunological conditioning with ¹³¹I-apamistamab (Iomab-B) opens up new possibilities for older patients with relapsed or refractory AML. Through targeted irradiation of CD45-positive cells, Iomab-B enables safe myeloablative conditioning. The phase III SIERRA study, reported in conference proceedings, showed that 100% of patients in the Iomab-B arm progressed to HCT and 22% achieved durable complete remission, while no durable remission was observed in the control arm [11]. As of 2025, Iomab-B has not yet received marketing authorization in Europe and is available only within clinical trials or early access/compassionate-use programs. This could enable more transplants for older high-risk patients in the future.

RIC vs MAC

In older or comorbid patients in particular, reduced conditioning lowers transplant-related mortality but comes at the cost of an increased risk of recurrence, as shown in randomized trials comparing RIC and MAC in AML/MDS [12]. Fit patients with MRD persistence or unfavorable genetics are advised to undergo myeloablative conditioning prior to allogeneic HCT in order to maximize engraftment and durable graft-versus-leukemia effects. In our center, commonly used RIC regimens include fludarabine with reduced-dose busulfan (Flu/Bu2) or fludarabine/melphalan, whereas MAC regimens typically consist of high-dose busulfan/fludarabine (Flu/Bu4) or cyclophosphamide/total body irradiation.

GvHD prophylaxis

PTCy in combination with tacrolimus and mycophenolate is now standard in many donor constellations. The randomized BMT CTN 1703 study demonstrated that PTCy + Tac/MMF significantly improves 1-year GVHD/relapse-free survival and reduces the incidence of severe acute GvHD [13]. Retrospective registry analyses show that PTCy achieves lower nonrelapse mortality and better overall survival than rATG in mismatched unrelated donors, while acute and

chronic GvHD remain comparable [7]. Abatacept is used as an add-on, particularly in 1-HLA-mismatched transplants [10]. Vedolizumab added to CNI-based prophylaxis (no PTCy) reduced lower-GI aGVHD or death at day +180 (85.5% vs 70.9%; HR 0.45; $p=0.0009$). Benefit was seen in lower-GI aGVHD-free and relapse-free survival, without OS difference [14].

Advances in acute and chronic GVHD management

Ruxolitinib, an oral JAK 1/2 inhibitor, has become the standard second-line treatment for steroid-refractory acute and chronic GVHD in many centers. In the phase III REACH2 trial in steroid-refractory aGVHD, ruxolitinib achieved significantly higher day-28 overall response rates and more durable responses compared with best available therapy [15]. In the phase III REACH3 trial in steroid-refractory or -dependent cGVHD, ruxolitinib improved week-24 overall response rates and markedly prolonged failure-free survival versus best available therapy, with sustained efficacy at long-term follow-up [16]. Accordingly, ruxolitinib is now approved for steroid-refractory aGVHD and cGVHD and is widely used in daily clinical practice.

Maintenance therapy

FLT3 inhibitors

In FLT3 ITD-mutated AML, sorafenib as maintenance therapy after allogeneic hematopoietic stem cell transplantation reduces the risk of relapse and prolongs survival, as previously shown [17]. Although the new MORPHO study with gilteritinib failed to meet its primary endpoint in the overall population, it showed a clear benefit for MRD-positive patients [18]. In a subsequent post hoc analysis, patients with concurrent *NPM1* mutation experienced the largest relapse-free survival benefit from post-HCT gilteritinib, and in this subgroup gilteritinib maintenance after RIC appeared as effective as after MAC in preventing relapse [19]. Addition of quizartinib, another FLT3 inhibitor, to induction, consolidation, and maintenance treatment (including patients consolidated with allo-HCT), resulted in an improved overall and relapse-free survival in patients with newly diagnosed AML with FLT3-ITD [20].

Hypomethylating agents and combinations

Azacitidine as maintenance showed no advantage over observation in a randomized phase III study after allo-HCT; no positive RCT data are available for decitabine [21]. HMA + venetoclax (e.g., Ven/Aza) shows encouraging survival rates in early phase I/II studies in selected patients with acceptable toxicity

[22]. The balance between efficacy and toxicity is crucial.

TP53-targeted therapy

In TP53-mutated AML, APR-246 (Eprexapopt) + azacitidine was investigated as maintenance therapy after allo-HCT. In a multicenter phase II study, the 1-year RFS was 59.9% (95% CI 41.0–74.4) and the 1-year OS was 78.8%; the median OS was 20.6 months. Although this mutation is still associated with a very high risk of relapse, such maintenance therapy could nevertheless improve survival [23], but this approach remains investigational and is not yet standard of care outside clinical trials.

IDH inhibitors: Initial reports on the post-HCT use of ivosidenib (IDH1) and enasidenib (IDH2) show sustained remissions and good tolerability in small cohorts [24, 25].

Menin inhibitors—most notably revumenib—are most relevant as a bridge to allo-HCT, with AUGMENT-101 demonstrating responses that enabled patients to proceed to transplant [26].

In current daily practice, FLT3 inhibitors (sorafenib and, where approved, gilteritinib or quizartinib), IDH1/2 inhibitors and venetoclax-based combinations are increasingly used in selected high-risk patients after allo-HCT, whereas APR-246 and menin inhibitors remain investigational and are currently restricted to clinical trials or early-access programs.

MRD monitoring and preemptive interventions

Sensitive MRD monitoring after HCT allows early detection of molecular relapse and opens a window of opportunity for preemptive measures. In a recent conference abstract, approximately 30% of patients were MRD-positive on day 30 after HCT, which was associated with significantly reduced relapse-free and overall survival [27]. Donor lymphocyte infusions are administered when MRD is detected or in cases of mixed chimerism to enhance the graft-versus-leukemia effect [28]. Studies show that early (pre-emptive) DLI can significantly reduce the risk of relapse; however, there is also an increased risk of GvHD [29]. Targeted therapies can also be used preemptively if a specific mutation is present (e.g., FLT3 inhibitors in FLT3-MRD). Overall, MRD-guided therapies will play an increasingly important role in post-HCT care.

Supportive therapy and the microbiome

Effective infection prophylaxis is crucial for transplant success. Letermovir reduces the rate of clinical CMV disease in real-world data to 1.3% by week 14 and 3.9% by week 48 and is established in CMV-seropositive recipients [30]. Antiviral options for ganciclovir-resistant CMV have been expanded with maribavir (phase III advantage over IAT). Antifun-

gal prophylaxis (e.g., posaconazole), pneumocystis prophylaxis (trimethoprim/sulfamethoxazole), and pneumococcal vaccinations remain the standard of supportive therapy [31].

The gut microbiome is emerging as a factor in the disease and treatment of GvHD and in the success of transplantation. High diversity of the gut flora prior to HCT correlates with better survival [32], and experimentally it is also associated with reduced gastrointestinal GvHD. Current randomized data (2025) show that FMT after allo-HCT is safe, effectively restores microbiome diversity, and—depending on the donor—provides evidence of a reduction in severe acute GvHD [33]. At EHA 2025, Early Access Program data on pooled allogeneic microbiotherapy reported favorable outcomes in steroid-refractory GI-aGvHD, supporting microbiome restoration as a therapeutic strategy [34].

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